

DAX Measures & Calculated Metrics Documentation

Project: Superstore Sales & Profit Dashboard — Week 3

Tool: Microsoft Power BI

#	Measure Name	DAX Formula	Purpose
1	Total Sales	Total Sales = SUM(superstore[Sales])	Calculates the sum of all sales across the dataset. Used as the primary revenue KPI and as the base for several other measures.
2	Total Profit	Total Profit = SUM(superstore[Profit])	Calculates the sum of all profit across the dataset. Used to assess overall business profitability.
3	Total Orders	Total Orders = DISTINCTCOUNT(superstore[Order ID])	Counts the number of unique orders, avoiding overcounting from multiple product line items within the same order.
4	Average Sales per Order	Average Sales per Order = DIVIDE([Total Sales], [Total Orders], 0)	Calculates the average value of a typical order, giving a more business-meaningful figure than averaging individual line items.
5	Profit Margin	Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)	Expresses profitability as a percentage of sales, allowing comparison across categories, regions, and time periods regardless of scale.
6	Average Discount	Average Discount = AVERAGE(superstore[Discount])	Calculates the average discount rate applied across transactions. Used to investigate the relationship between discounting and profitability (Business Problem 3).

Notes on measure design choices:

- Total Orders uses DISTINCTCOUNT rather than COUNT because the Superstore dataset stores one row per product line item, not one row per order, a single order can span multiple rows.
- Average Sales per Order was deliberately built using DIVIDE(Total Sales, Total Orders) rather than AVERAGE(Sales), since averaging the raw Sales column would return the average *line-item* value rather than the average *order* value, a less useful figure for business decision-making.
- DIVIDE() is used instead of the / operator throughout to safely handle potential divide-by-zero errors (it returns 0 or a specified alternate value instead of an error).